A Modular Computational Framework for Section 2 Voting Rights Act Analysis

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Abstract

A modular computational framework for §2 Voting Rights Act is introduced which integrates geospatial data engineering, Bayesian ecological inference, and ensemble generation. Prior work has developed these components individually, existing approaches lack a unified pipeline that propagates uncertainty from voter behavior estimation through plan-level evaluation. A probabilistic notion of minority electoral opportunity is formalized. A case study of Texas congressional districts illustrates how the framework can be used to assess whether an enacted plan systematically dilutes minority voting power.

Introduction

Gerrymandering

Gerrymandering is a tactic that has been interwoven with United States politics since at least 1812, which is when the term was first coined [Sauer, 1992]. In simplest terms, gerrymandering can be defined as the manipulation of election district maps to either favor or disfavor a group of people (i.e. a political party, a racial/ethnic group, a socioeconomic class, etc).

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Negative Racial Gerrymandering

Negative racial gerrymandering dilutes minority voting power by engineering districts in which minority ballots systematically matter less. Although minority voters retain the formal right to vote, they lack proportional influence.

Unfortunately, this practice has materialized throughout U.S. history. For instance, in response to black Americans obtaining the right to vote in 1870, Reconstruction-era Redeemer Southern Democrats wielded negative racial gerrymandering to preserve white supremacy and curtail black political power [Valelly, 2004].

This distortion produces legislatures that are less representative—often whiter and more ideologically extreme than the electorate—and weakens elected officials' incentives to respond to minority interests.

By breaking the link between voting and representation, negative racial gerrymandering undermines democratic legitimacy, erodes trust, and contributes to lower turnout and political disengagement.

The Legality of Negative Racial Gerrymandering

Negative racial gerrymandering is illegal in the United States by virtue of the Equal Protection Clause of the Fourteenth Amendment and Section 2 of the Voting Rights Act.

The Equal Protection Clause of the Fourteenth Amendment forbids denying anyone of "equal protections of the law" [United States Congress, 1868].

Section 2 of the Voting Rights Act of 1965 precludes states and political entities from wielding any "standard, practice, or procedure" to curtail the voting rights of any U.S. citizen based on race [United States Congress, 1965]. In the 1986 case of Thornburg v. Gingles, it was ruled that a North Carolina redistricting plan was in violation of Section 2 of the Voting Rights Act for engaging in negative racial gerrymandering practices - more specifically, of diluting the votes of black citizens [Supreme Court of the United States, 1986]. This landmark Supreme Court case decided that minority voters can legally challenge negative racial gerrymandering practices in redistricting plans on the grounds of violating Section 2 of the Voting Rights Act, but that the burden of proof is on them to demonstrate that their capacity to elect their preferred candidates is undermined [Supreme Court of the United States, 1986].

In the Thornburg v. Gingles case, the Supreme Court established a framework for minority voters to be able to prove negative racial gerrymandering in violation of Section 2 of the Voting Rights Act in a court of law [Supreme Court of the United States, 1986]. Under this framework, plantiffs must be able to prove the following: (1) a "geographically insular" minority group makes up the majority population of a district, (2) the minority group is "politically cohesive", and (3) the "bloc voting majority" is usually able to defeat the minority group's preferred candidates [Supreme Court of the United States, 1986]. The second and third preconditions are stated by the U.S. Supreme Court as concerning "racially polarized voting" [Supreme Court of the United States, 2023].

The U.S. Supreme Court refers to this framework as "the Gingles framework" [Supreme Court of the United States, 2023]. For the last 40 years, claims of negative racial gerrymandering in violation of Section 2 of the Voting Rights Act have been assessed via the Gingles framework [Supreme Court of the United States, 2023]. In the 2023 Allen v. Milligan decision, the U.S. Supreme Court reaffirmed the validity of the Gingles framework [Supreme Court of the United States, 2023].

Related Work and Contribution

Redistricting Data Preparation

Constructing Racial Categories

In this paper, the Data and Democracy Lab's conventions for constructing demographic categories are adopted.

Their convention for categorizing racial groups involves adopting these group names in the ensuing order: Black, Hispanic, Asian, Other,and White [Donnay, M. and Duchin, M. and Rock, P., n.d.].

It is emphasized that anyone who identifies as black should be grouped into the black population [Donnay, M. and Duchin, M. and Rock, P., n.d.]. In other words, the black population should encompass those who are "Any-Part Black" or "Black Alone or in Combination", as opposed to just "Black Alone" or "Single-Race Black" [Donnay, M. and Duchin, M. and Rock, P., n.d.].

Then, of the individuals left, the Hispanic population must encompass all individuals who identify as Hispanic in the same way the Black population does [Donnay, M. and Duchin, M. and Rock, P., n.d.]. Next, of those left, the Asian population should capture all who identify as Asian or Native Hawaiin Pacific Islander, so as to represent an AAPI category [Donnay, M. and Duchin, M. and Rock, P., n.d.]. Finally, the same logic is used to construct the AMIN, Other, and White categories [Donnay, M. and Duchin, M. and Rock, P., n.d.].

Thereafter, source data:pl94 in Section sec:datasources is leveraged to generate voting age population as sums of census variables [Donnay, M. and Duchin, M. and Rock, P., n.d.].

Subsequently, Source **??** in §@ref(datasources) is analyzed. The problem is, CVAP data from the ACS must be analyzed at the census block level because the base unit of the Decennial Census PL 94-171 data is the census block. In the primary tables, ACS is reported at the census tract level, which presents a problem because that is a significantly higher level of aggregation than the census block [Donnay, M. and Duchin, M. and Rock, P., n.d.].

Fortunately, the Data and Democracy Lab has introduced a solution to this problem called "discounting" [Donnay, M. and Duchin, M. and Rock, P., n.d.]. Discounting refers to the usage of the tract-level ACS data to estimate the citizenship rates of racial groups, then applying the discount to the voting age population [Donnay, M. and Duchin, M. and Rock, P., n.d.].

An example of "discounting" provided by the Data and Democracy Lab is depicted in Equation~**??**; in this example, block X belongs to tract Y in state Z and BCVAP is estimated by taking the citizenship rate for the group in the tract and applying it to the voting ager population in the block [Donnay, M. and Duchin, M. and Rock, P., n.d.].

$$\frac{BCVAP_{ACS}(tract\,Y)}{BVAP_{ACS}(tract\,Y)} \times BVAP_{PL}(block\,X) = BCVAP(block\,X)$$

If the denominator in an equation like Equation~**??** is too low, then the Data and Democracy Lab proposes a solution like that in Equation~**??**. This involves swapping in the citizenship rate given that the tract did not allow for a meaningful estimate [Donnay, M. and Duchin, M. and Rock, P., n.d.].

$$\frac{BCVAP_{ACS}(state\,Z)}{BVAP_{ACS}(state\,Z)} \times BVAP_{PL}(block\,X) = BCVAP(block\,X)$$

The Data and Democracy Lab acknowledges that this is a "crude fix", but states that they intend to release a "more sophisticated" Bayesian method to address this in the future [Donnay, M. and Duchin, M. and Rock, P., n.d.].

Ecological Inference

To reiterate what is stated in §@ref(legality), in order to prove a violation of §2 of the Voting Rights Act under the Gingles framework,plaintiffs must prove that the majority votes as a bloc to defeat minority-preferred candidates [Supreme Court of the United States, 1986]. Hence, any framework performing §2 Voting Rights Act analysis must encompass identification of minority-preferred political candidates [Becker et al., 2021].

In state election return datasets, vote totals for political candidates are not computed by race, meaning that direct computation of minority-preferred candidates is impossible [Becker et al., 2021]. This is where ecological inference - the de-facto solution for determining racially polarized voting - comes in. After all, ecological inference is widely regarded as "the best and often only hope of making progress" when it comes to extracting clues about individual behavior from group or aggregate-level data [King et al., 2004].

Ecological inference (and ecological regression, its simpler predecessor) is considered a "statistical mainstay" in §2 Voting Rights Act litigation [Freedman, 1999]. U.S. Supreme Court redistricting cases in which lower-court findings relying on ecological inference were reviewed and accepted without methodological objection include Allen v. Milligan [Supreme Court of the United States, 2023], Alexander v. State Carolina State Conference of the NAACP [Supreme Court of the United States, 2024], Cooper v. Harris [Supreme Court of the United States, 2017a], Bethune-Hill v. Virginia State Board of Elections [Supreme Court of the United States, 2017b], and Abbott v. Perez [of the United States, 2018].

It should be noted, however, that in the same way all statistical inference techniques with missing data give way to uncertainty, ecological inference produces "imperfect" and "uncertain" answers [Becker et al., 2021]. Hence, it is essential to estimate the error produced by ecological inference techniques and observe how it compounds or cancels out in high-level conclusions [Becker et al., 2021].So, in addition to identifying the minority-preferred candidates, one must specify the confidence measures that the identifications are correct [Becker et al., 2021]. To check for minority control of a district and identify preferred candidates for newly proposed districts, the statewide and precinct-level vote estimates by race should be leveraged [Becker et al., 2021].

Becker et al. [Becker et al., 2021] implement a version of King's ecological inference, specifically the `ei.mD.bayes` function from eiPack which is based on the Bayesian hierarchical Dirichlet model for $R \times C$ tables proposed by King et al. in 1999 [King et al., 1999].

For each election, EI is run at the statewide level using precinct-level input tables [Becker et al., 2021]. The inputs for each precinct are the row and column sums for the $R \times C$ table of vote counts. The row sums correspond to the precinct's estimated number of adult citizens in each racial group. The column sums are the precinct's vote totals for each candidate [Becker et al., 2021].

Each ecological inference run generates a large random sample of estimated vote counts; in order to obtain the statewide estimates, these values are summed across the entire state [Becker et al., 2021]. For each racial group, the candidate with the highest average estimated vote total for an election is the preferred candidate [Becker et al., 2021]. In order to obtain a measure of confidence for the a candidate was correctly identified as the preferred candidate, repeated draws for the ecological inference distribution are obtained and the frequency by which the preferred candidate receives the most votes is recorded [Becker et al., 2021].

Ensemble-Based Outlier Analysis in Redistricting

Recombination Markov Chains

The early ensembles relied on flip-based Markov chains, which propose single-precinct boundary changes between adjacent districts. Flip chains have well-understood detailed balance with respect to a known target measure but mix poorly on real precinct graphs: the proposal kernel changes the partition only marginally per step, and the chain remains correlated with its starting state for chain lengths far beyond what is computationally feasible.

In 2021, Daryl DeFord, Moon Duchin, and Justin Solomon set forth recombination, a novel family of Markov chains intended for the ensemble generation of alternative redistricting plans [DeFord et al., 2021]. Recombination is widely considered to be the standard engine for district plan ensemble generation in the redistricting space [Becker et al., 2021].

— which proposes whole-district swaps via uniform spanning-tree bipartition of merged adjacent districts — represented a substantial advance in proposal mobility and is now the field's default kernel. The MGGG case studies on Virginia, Massachusetts, and Pennsylvania (Duchin et al., 2018, 2019) and the GerryChain software library that emerged from this work have established ReCom's position as the standard tool for redistricting ensemble generation.

The Mixing-Time Problem

The Framework

Data Engineering Layer

Data Sources

1. State District Plan Shapefile
2. Decennial Census PL 94-171 Redistricting Dataset
3. Citizen Voting Age Population by Race and Ethnicity in a Special Tabulation from the ACS 5-Year Estimates
4. State Census Blocks Shapefile (TIGER/LINE)
5. State Voting Tabulation District Shapefile
6. State VTDs Election Data

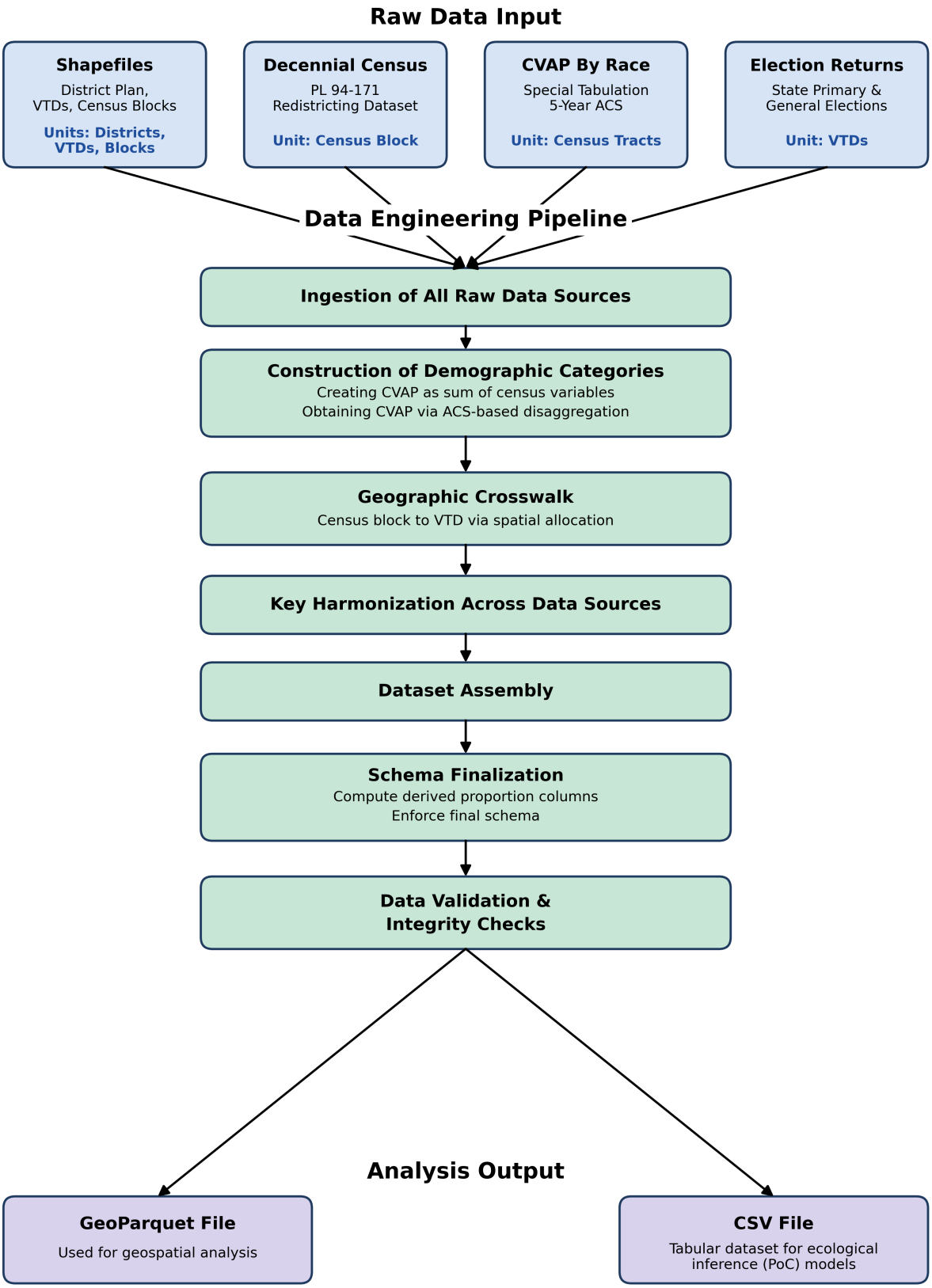


Figure 1: Overview of the data engineering layer.

Ecological Inference Layer

The ecological inference layer's first inferential stage converts observed precinct-level election returns and citizen voting-age population (CVAP) composition into group-conditional posterior distributions over vote choice. These distributions are the substrate on which all downstream Voting Rights Act opportunity scoring is built: they identify the minority-preferred candidates in each contest, they provide precinct-level posterior summaries of group-specific vote counts (used to construct district-level support estimates over the ensemble of plans), and they supply the raw scores to which a per-mode, per-group logistic calibration is fit.

Inputs and the Inferential Target

The dataset generated in the data engineering layer is a precinct-level table covering all of Texas. For each precinct p there is a vector of CVAP shares $X_p = (X_{p,1}, \ldots, X_{p,R}) \in \Delta^{R-1}$ across $R = 5$ racial groups (Hispanic, Black, White, Asian, and American Indian/Native) together with per-candidate vote counts and total turnout for each contest in our election inventory. Mixed primaries are split by party so each fitted election has a single, party-specific non-voting category; the Republican "Uncommitted" line in the 2024 Presidential primary is dropped because it admits no candidate-level racial identification.

Because the secret ballot makes the joint distribution of race and vote choice unobservable, we recover it by ecological inference: a model that reconciles the two observed marginals — racial composition by precinct and vote totals by precinct — to produce a credible posterior over within-precinct, group-specific support.

Model and Inference

We use the row-by-column (RxC) hierarchical Multinomial-Dirichlet Bayesian model of **?**, the standard specification for redistricting EI since **?**. Let C index the columns: one per candidate plus an *Abstain* column, so column totals sum to CVAP rather than to ballots cast. For each precinct p and each group r, define a latent simplex $\beta_{p,r,\cdot} \in \Delta^{C-1}$ giving group r's probabilities over the C outcomes in precinct p. The hierarchical specification is

$$\beta_{p,r,\cdot} \mid \alpha_r \;\sim\; \text{Dirichlet}(\alpha_r), \qquad \alpha_r \;\sim\; \pi(\alpha_r), \qquad r = 1, \ldots, R,\;\; p = 1, \ldots, P,$$

where $\pi(\alpha_r)$ is the weakly-informative hyperprior implemented by the `RowByColumnEI` class of `pyei` [**?**]. Conditional on $\{\beta_{p,r,c}\}$ and the (fixed) group fractions X_p, the precinct-level outcome probabilities are the group-share-weighted mixture

$$\theta_{p,c} \;=\; \sum_{r=1}^{R} X_{p,r}\, \beta_{p,r,c}, \qquad c = 1, \ldots, C,$$

and the observed column counts $V_{p,\cdot}$ — votes by candidate plus the abstention residual — follow

$$V_{p,\cdot} \mid \theta_p \;\sim\; \text{Multinomial}(N_p, \theta_p), \qquad N_p \;=\; \text{CVAP}_p.$$

Treating CVAP as the multinomial denominator (rather than ballots cast) is deliberate: it ties the abstention rate to the same demographic composition used to identify candidate support, so heterogeneity in turnout across groups is absorbed into the posterior over β rather than being silently projected out by re-normalising the columns. For each contest we fit (4.1)–(4.3) independently; precincts with $N_p = 0$ or with no votes recorded for the contest are excluded, and contests retaining fewer than 50 precincts after filtering are dropped.

Posterior inference uses the No-U-Turn Sampler [**?**] with 1{,}000 retained draws per chain after 100{,}000 tuning steps, a single chain, and a fixed random seed; sampling is performed in single precision to keep the posterior tensor (draws $\times$ precincts $\times$ groups $\times$ candidates) within memory. Per-contest traces are checkpointed to NetCDF so that the postprocessing stage can be rerun without re-sampling.

Posterior Summaries

Three quantities are exported per contest. Let S index the retained MCMC draws and let $\beta^{(s)}_{p,r,c}$ denote the s-th draw.

Statewide preferences. Group r's statewide support for candidate c in draw s is the CVAP-weighted average of within-precinct probabilities,

$$\hat{S}^{(s)}_{r,c} \;=\; \frac{\sum_p X_{p,r}\, N_p\, \beta^{(s)}_{p,r,c}}{\sum_p X_{p,r}\, N_p}.$$

The candidate of choice is $c^\star_r = \arg\max_c \overline{S}_{r,c}$ where $\overline{S}_{r,c} = S^{-1}\sum_s \hat{S}^{(s)}_{r,c}$. We report $\overline{S}_{r,c^\star_r}$ alongside the fraction of draws under which $c^\star_r$ is the modal preference,

$$\pi_r \;=\; \frac{1}{S}\sum_{s=1}^{S} \mathbf{1}\Big[\arg\max_c \hat{S}^{(s)}_{r,c} \;=\; c^\star_r\Big].$$

The latter quantity, written `prob` in the downstream tables, supplies the posterior credibility that group r's preferred candidate is in fact $c^\star_r$ and is used to weight contests in the cohesion calculation.

Precinct mean counts. For each (precinct, group, candidate) we record the posterior mean estimated count

$$\overline{n}_{p,r,c} \;=\; X_{p,r}\, N_p\, \frac{1}{S}\sum_{s=1}^{S} \beta^{(s)}_{p,r,c}.$$

Precinct count octiles. Within-precinct posterior uncertainty is preserved for downstream Monte Carlo by exporting eight octile quantiles of the group-specific count posterior, evaluated at the probabilities $\mathcal{Q} = \{k/8 : k = 1, \ldots, 8\}$:

$$Q_{p,r,c}(q) \;=\; \mathrm{Quantile}_q\Big(\big\{X_{p,r}\, N_p\, \beta^{(s)}_{p,r,c}\big\}_{s=1}^{S}\Big), \qquad q \in \mathcal{Q}.$$

The ensemble layer treats each successive pair of octiles as a uniform bin and samples linearly within it, recovering an empirical eight-bin approximation to the full posterior distribution of $X_{p,r}N_p\beta_{p,r,c}$. The implicit floor $q = 0$ is set to zero, an approximation that biases the simulated draw floor downward only when the posterior places non-negligible mass near zero counts.

These three artefacts are pivoted from the long, draw-aware format produced by the sampler into the wide, composite-key format `{CVAP\_KEY}.{election}\_{candidate}.{permille}` consumed by the ensemble layer, where `CVAP_KEY` maps the EI group label to the corresponding CVAP column (Hispanic $\rightarrow$ HCVAP, Black $\rightarrow$ BCVAP) and `permille` is the integer $\lfloor 1000 \cdot q \rfloor$. We additionally emit a single-row CSV of in-group cohesion weights $w \in \{1.0, 0.75, 0.5\}$ that the ensemble layer applies to elections in which the candidate of choice is, respectively, a co-ethnic of the group, a co-ethnic of one of the two minority groups but not both, or neither.

Calibration of VRA-Opportunity Scores

The ensemble-layer scoring function (Section 5) returns, for each district d, each model mode $m \in \{\text{statewide}, \text{equal}, \text{district}\}$, and each subgroup $g \in \{\text{Black}, \text{Latino}, \text{Neither}\}$, a raw VRA-opportunity score $\rho_{d,m,g} \in [0, 1]$ constructed as a weighted share of minority-preferred-candidate votes. Raw scores are internally consistent across plans but are not directly interpretable as probabilities of the corresponding

opportunity outcome; their ordinal information must be mapped to a calibrated probability scale before being aggregated across modes and subgroups via the Venn-overlap formula used in Section 5.

We calibrate $\rho_{d,m,g}$ by fitting one univariate logistic regression per (m, g) cell — nine models in total. The training data are produced by scoring the enacted Texas congressional plan with the ensemble-layer machinery in `return_raw` mode, harvesting $\{\rho_{d,m,g}\}_{d=1}^{38}$, and pairing each district with a binary label $y_{d,m,g}$ derived from a benchmark general-election outcome:

$$y_{d,m,\text{Black}} \;=\; \mathbf{1}[\text{black-preferred candidate wins district } d]\,,$$

with $y_{d,m,\text{Latino}}$ defined analogously and $y_{d,m,\text{Neither}} = 1 - \max(y_{d,m,\text{Black}}, y_{d,m,\text{Latino}})$. Multiple benchmarks may be pooled; raw scores depend only on the enacted plan and so are computed once and reused across benchmarks. For each (m, g) we then fit

$$\Pr(y_{d,m,g} = 1 \mid \rho_{d,m,g}) \;=\; \sigma(\alpha_{m,g} + \beta_{m,g}\,\rho_{d,m,g})\,, \qquad \sigma(z) = \frac{1}{1+e^{-z}},$$

via Newton-Raphson on the penalised log-likelihood

$$\ell(\alpha, \beta) \;=\; \sum_d \Big[y_d \log \sigma(\alpha + \beta\rho_d) + (1 - y_d) \log\big(1 - \sigma(\alpha + \beta\rho_d)\big) \Big] - \tfrac{\lambda}{2}(\alpha^2 + \beta^2),$$

with $\lambda = 10^{-6}$ for numerical stability. When a calibration cell contains only one observed class (a common occurrence for the Neither subgroup under tightly clustered benchmarks) we revert to the identity calibration $\alpha_{m,g} = 0$, $\beta_{m,g} = 1$, propagating the raw score unchanged rather than allowing complete separation to drive $|\beta|$ to infinity.

The fitted parameters $\{(\alpha_{m,g}, \beta_{m,g})\}$ are written to a nine-row table that the ensemble layer consults whenever it requires a calibrated opportunity probability. Recalibration is inexpensive, requires no re-sampling of the EI posterior, and is the natural point at which to ingest new benchmark data when additional general elections become available.

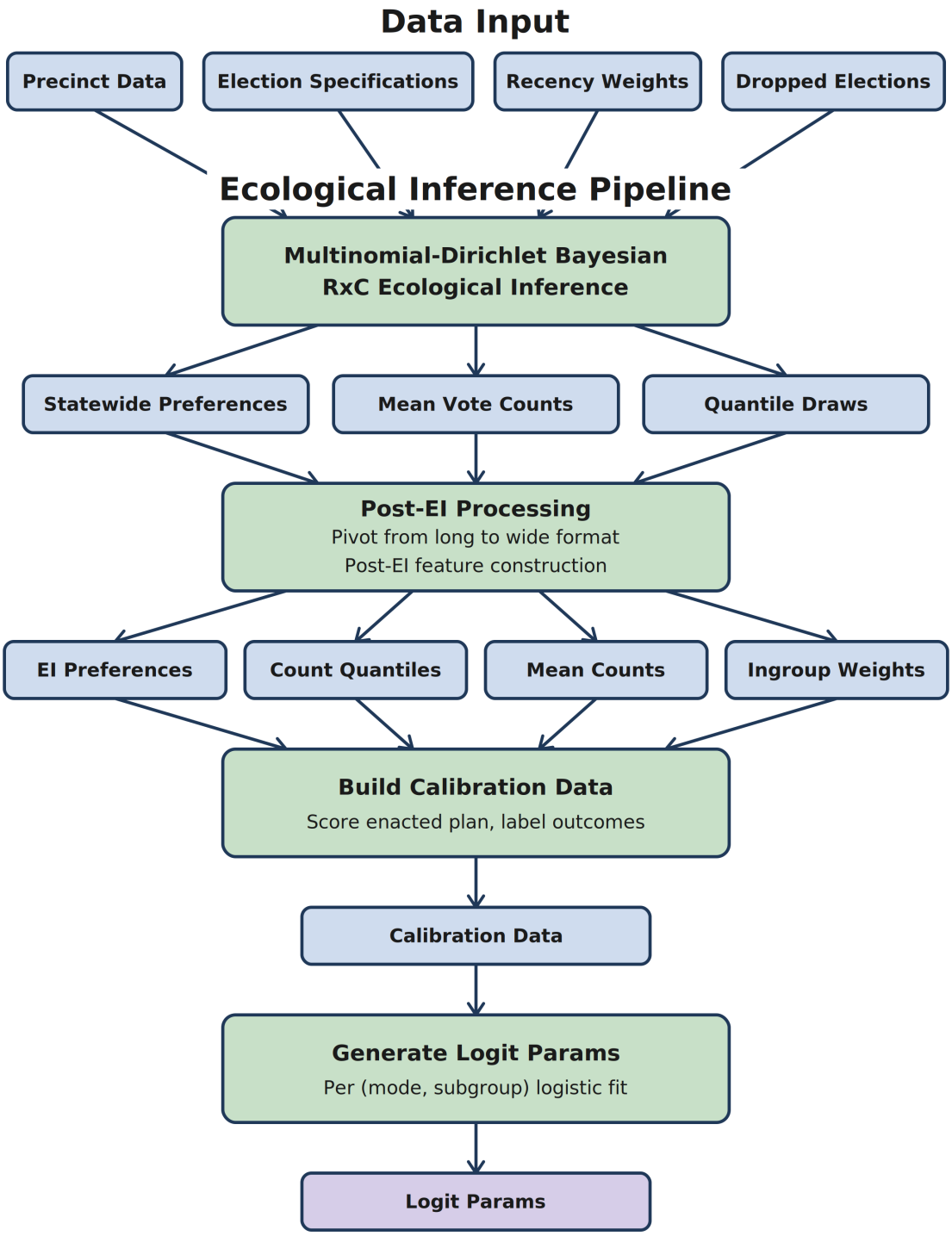


Figure 2: Overview of the ecological inference layer.

Ensemble Generation & Statistical Comparison Layer

Setup and Notation

Let $V = \{v_1, \ldots, v_N\}$ denote the precincts in the state, with rook-adjacency graph $G = (V, E)$. A *plan* is a partition $\pi : V \to \{1, \ldots, K\}$ of V into K districts; the enacted Texas congressional plan ($K = 38$) is denoted π_0. Let $\Pi_K(\varepsilon)$ denote the set of plans for which (i) every district induces a connected subgraph, and (ii) the population-balance condition $|\mathrm{pop}(d)/(\mathrm{pop}_{\mathrm{tot}}/K) - 1| \le \varepsilon$ holds for every d. We work throughout with $\varepsilon = 0.05$, the slack at which π_0 itself is a valid initial state under the precinct-vintage population column used here.

Let ρ denote the *spanning-tree measure* on $\Pi_K(\varepsilon)$,

$$\rho(\pi) \;\propto\; \prod_{d=1}^{K} \tau\big(G[\pi^{-1}(d)]\big),$$

where $\tau(H)$ is the number of spanning trees of subgraph H. This is the canonical neutral measure targeted by Recombination (ReCom) Markov chains [DeFord, Duchin, and Solomon, 2021; Cannon et al., 2022]. The hypothesis we test is

$$H_0 : \pi_0 \sim \rho_{\mathcal{C}},$$

where $\rho_{\mathcal{C}}$ is ρ restricted to a constraint subset $\mathcal{C} \subseteq \Pi_K(\varepsilon)$ encoding compactness and county-respect requirements (defined in §4.3). The test statistics are scalar functionals $T : \Pi_K(\varepsilon) \to \mathbb{R}$ measuring minority opportunity.

The mixing time of practical ReCom kernels on real-world precinct graphs is unknown and conjecturally polynomial but slow. The inference procedure adopted below sidesteps this concern: validity holds *exactly*, regardless of mixing.

The Opportunity Functional

For each protected group $g \in \{B, L\}$ (Black, Latino), each plan π, and each district d, let $p_d^g(\pi) \in [0, 1]$ denote the model-based probability that group g's preferred candidate prevails in district d. The probabilities $\{p_d^g(\pi)\}$ are produced by the scoring layer detailed in §3, which combines a row-by-column ecological-inference model of group-specific candidate preferences with recency-weighted election histories under three weighting schemes (statewide, equal, district). Primary results below use the *statewide* mode; results under the alternative modes are reported in Appendix [X].

We define the *canonical opportunity functional* as the sum of district-level win probabilities,

$$O_g(\pi) \;=\; \sum_{d=1}^{K} p_d^g(\pi),$$

and a coalition functional crediting each district to whichever group has the higher win probability,

$$O_{\mathrm{joint}}(\pi) \;=\; \sum_{d=1}^{K} \max\big\{p_d^B(\pi),\, p_d^L(\pi)\big\}.$$

$O_g(\pi)$ admits an axiomatic justification (anonymity, district-additivity, monotonicity, normalization) given in §2. As an estimand it is the expected number of districts in which group g's preferred candidate prevails, given the win probabilities supplied by the scoring layer.

To express deviation from the *De Grandy* proportional-representation baseline [*Johnson v. De Grandy*, 512 U.S. 997 (1994)], let $\bar{m}_g$ denote the statewide CVAP share of group g, and define the diversity-calibrated functional

$$F_g(\pi) \;=\; O_g(\pi) - K\,\bar{m}_g,$$

with F_{joint} defined analogously using $\bar{m}_B + \bar{m}_L$. F_g is a translation of O_g that centers the demographic baseline at zero; $F_g(\pi) < 0$ indicates expected opportunity below proportionality. We adopt F_g as the primary test statistic, and report O_g and the ensemble-standardized

$$Z_g(\pi) \;=\; \big(O_g(\pi) - \mathbb{E}_{\rho_{\mathcal{C}}}[O_g]\big)\big/\sqrt{\mathrm{Var}_{\rho_{\mathcal{C}}}(O_g)}$$

as auxiliary quantities, with $\mathbb{E}_{\rho_{\mathcal{C}}}$ and $\mathrm{Var}_{\rho_{\mathcal{C}}}$ estimated from the spoke ensemble of §4.4.

The Neutral Ensemble: ReCom with VRA-Relevant Constraints

Proposals are generated by ReCom [DeFord, Duchin, and Solomon, 2021]. At each step the kernel (a) selects an adjacent district pair uniformly at random, (b) draws a uniform spanning tree on the merged subgraph, and (c) selects an edge whose removal yields a balanced bipartition. Combined with always-accept and the *allow-pair-reselection* variant of `bipartition_tree`—which re-rolls the district pair when no balanced cut exists rather than rejecting the proposal outright—the kernel P approximately satisfies detailed balance with respect to ρ on the constrained state space [DeFord, Duchin, and Solomon, 2021; Cannon et al., 2022]. We rely below on this approximate reversibility; see §4.6 for a discussion of the residual exactness gap.

The constraint set $\mathcal{C}$ comprises the population-balance requirement embedded in $\Pi_K(\varepsilon)$ together with two state-relative thresholds:

(i) *Compactness.* Total cut-edges, $\mathrm{CE}(\pi) = |\{e \in E : \pi(u) \neq \pi(w) \text{ for } e = \{u, w\}\}|$, must satisfy

$$\mathrm{CE}(\pi) \;\leq\; s_c \cdot \mathrm{CE}(\pi_0), \qquad s_c = 1.5.$$

The slack s_c is conventional in the redistricting literature [Mattingly et al., e.g., Herschlag et al., 2020]. Setting $s_c = \infty$ recovers ReCom's implicit spanning-tree compactness preference alone.

(ii) *County splits.* Let $\mathrm{CS}(\pi)$ denote the number of counties whose precincts span two or more districts. We impose

$$\mathrm{CS}(\pi) \;\leq\; s_g \cdot \mathrm{CS}(\pi_0), \qquad s_g = 1.2.$$

This constraint addresses the standard expert-witness objection that ensemble plans achieve elevated minority-opportunity scores by carving counties more aggressively than feasible legislative drafts. The 20% slack is chosen to preserve the constraint's bite while admitting sufficient mobility for the chain to mix; tighter slacks substantially slow mixing on the Texas precinct graph.

Both thresholds are scaled to π_0 rather than imposed in absolute terms, rendering $\mathcal{C}$ state-relative: the null distribution we compare π_0 against consists of plans that are no less compact and no more county-splitting than π_0 itself, scaled by the slacks.

We do *not* impose a hard contiguity validator. ReCom's spanning-tree mechanism guarantees that every accepted proposal is contiguous by construction; imposing contiguity additionally as a validator would cause the chain to reject π_0 as initial state, because the within-district subgraphs of certain enacted Texas CDs are not graph-connected under the precinct-vintage rook adjacency. This is an artifact of urban CD geometry interacting with rook adjacency rather than a substantive discontiguity.

Besag-Clifford Exact Inference

Conventional ReCom analyses report tail-rank statistics from a single long chain and rely on convergence diagnostics to defend their validity. We instead apply the *parallel method* of Besag and Clifford [1989, Proposition 3.3], which yields exact p-values under H_0 regardless of the kernel's mixing time.

Construction. Starting from π_0, run P forward for L_{hub} steps to obtain a *hub* state

$$X^* \;\sim\; P^{L_{\text{hub}}}(\pi_0, \cdot).$$

From X^*, generate M conditionally independent *spokes*

$$Y^{(1)}, \ldots, Y^{(M)} \overset{\text{i.i.d.}}{\sim} P^{L_{\text{spoke}}}(X^*, \cdot).$$

Under H_0 and the reversibility of P with respect to $\rho_{\mathcal{C}}$, the joint law of (π_0, X^*) is identical to that of (X^*, π_0), so π_0 given X^* has the same distribution as each $Y^{(j)}$ given X^*. Consequently $(\pi_0, Y^{(1)}, \ldots, Y^{(M)})$ is exchangeable conditional on X^*, and the rank of any test statistic T evaluated at π_0 within this $(M+1)$-tuple is uniform on $\{1, \ldots, M+1\}$.

The lower-tail Besag-Clifford p-value is therefore

$$\widehat{p}_{\text{low}}(T) \;=\; \frac{\left|\{j : T(Y^{(j)}) \leq T(\pi_0)\}\right| + 1}{M+1}, \tag{5}$$

and satisfies $\Pr_{H_0}\big(\widehat{p}_{\text{low}}(T) \leq \alpha\big) \leq \alpha$ for every $\alpha \in \{1/(M{+}1), 2/(M{+}1), \ldots\}$. The upper-tail expression is analogous. Crucially, the argument requires only reversibility of P with respect to $\rho_{\mathcal{C}}$; it places no requirement on L_{hub} or L_{spoke}.

Hyperparameter selection. The validity of (5) is independent of L_{hub}, L_{spoke}, and M, but each affects either resolution or power.

Resolution. The minimum attainable lower-tail p-value under (5) is $1/(M+1)$. We adopt $M = 1000$, giving a floor of $\approx 10^{-3}$ — the modal choice in published redistricting work [Mattingly group; MGGG case studies]. Raising M to 10,000 (floor $\approx 10^{-4}$) is appropriate for litigation/expert-witness contexts and is supported by the implementation without modification beyond compute.

Power. L_{hub} and L_{spoke} govern dependence structure within the spoke ensemble. Small L_{hub} leaves X^* and hence the spokes correlated with π_0, depressing power against alternatives near π_0 in the state graph; small L_{spoke} leaves spokes correlated with one another, inflating estimator variance for $\mathbb{E}_{\rho_{\mathcal{C}}}[O_g]$ and $\text{Var}_{\rho_{\mathcal{C}}}(O_g)$. We take $L_{\text{hub}} = 2000$ and $L_{\text{spoke}} = 500$. The asymmetric ratio reflects that X^* is generated once and shared across all spokes (so the unit cost of L_{hub} is amortized) while the per-spoke cost scales linearly in L_{spoke}.

Case Study: Texas Congressional Districts

Data Engineering Layer Implementation

Data Sources

1. Texas PLANC2333 District Plan Shapefile
2. Decennial Census PL 94-171 Redistricting Dataset
3. Citizen Voting Age Population by Race and Ethnicity in a Special Tabulation from the ACS 5-Year Estimates
4. Texas Census Blocks Shapefile (TIGER/LINE)
5. Texas Voting Tabulation District Shapefile
6. State VTDs Election Data

Table : Texas Demographics

Racial Group	Share of Total Population	Share of VAP	Share of CVAP
Hispanic	34.7%	35.6%	32.6%
Black	13.5%	13.0%	12.9%
White	46.2%	43.2%	48.9%
Asian	5.0%	6.1%	4.8%
American Indian	0.7%	1.3%	0.7%

Racial Group	Share of Total Population	Share of VAP	Share of CVAP

Table : Texas Statewide Elections Used

Office	Stage	Year
President	General	2024
President	Primary	2024
U.S. Senate	General	2024
U.S. Senate	Primary	2024
Railroad Commissioner	General	2024
Railroad Commissioner	Primary	2024

Ecological Inference Layer Implementation

Figure depicts the highest and lowest ecological inference point estimates of each racial group's support for the Democratic nominee across 3 Texas-statewide 2024 general elections (President,U.S. Senate, and Railroad Commissioner).

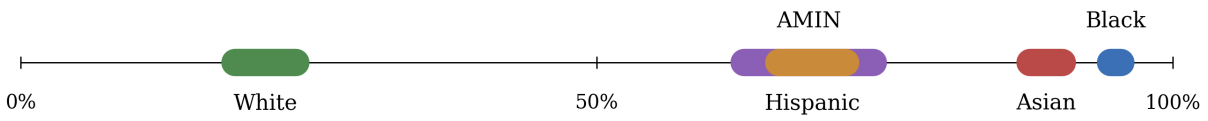


Figure 3: The highest and lowest EI point estimates for each racial group's support of the 3 Democratic nominees in statewide general elections: White (19–24%), AMIN (63–74%), Hispanic (66–72%), Asian (88–90%), and Black (95–96%).

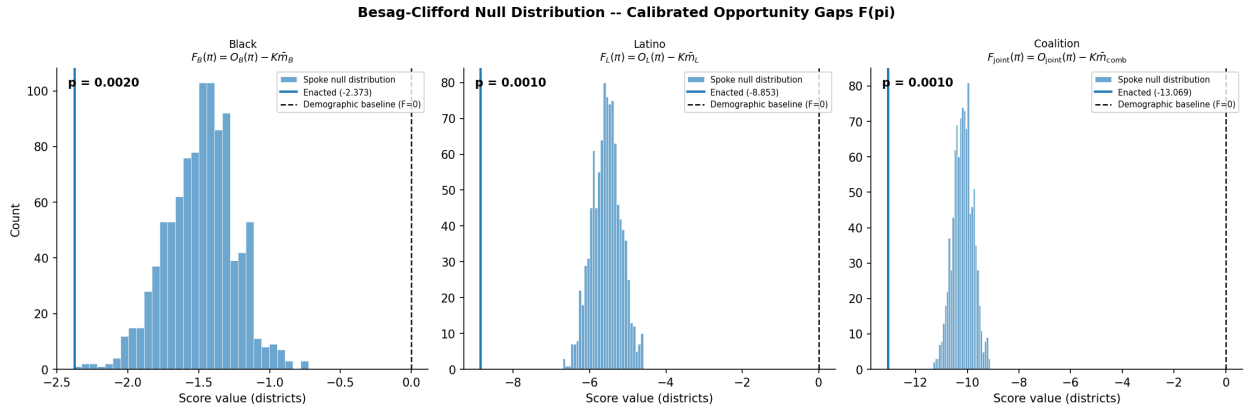


Figure 4: f distributions

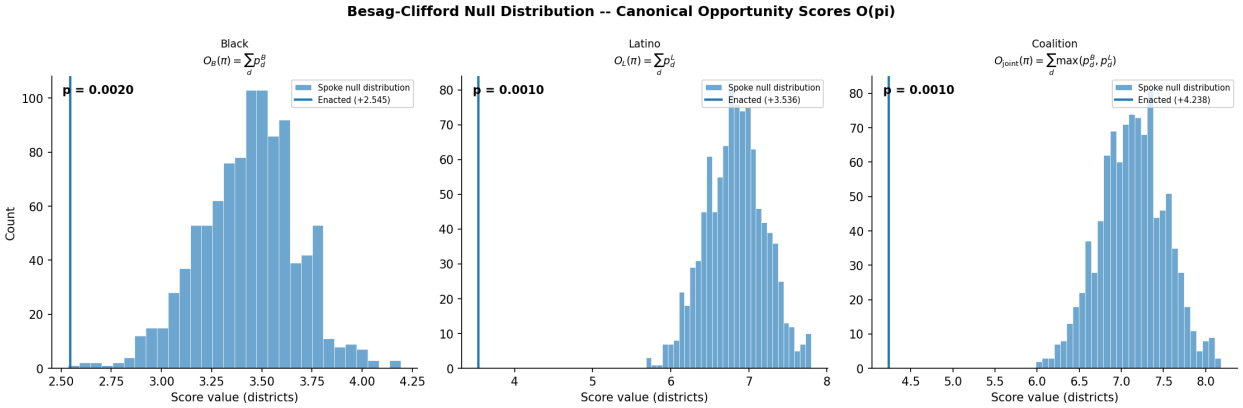


Figure 5: O dist

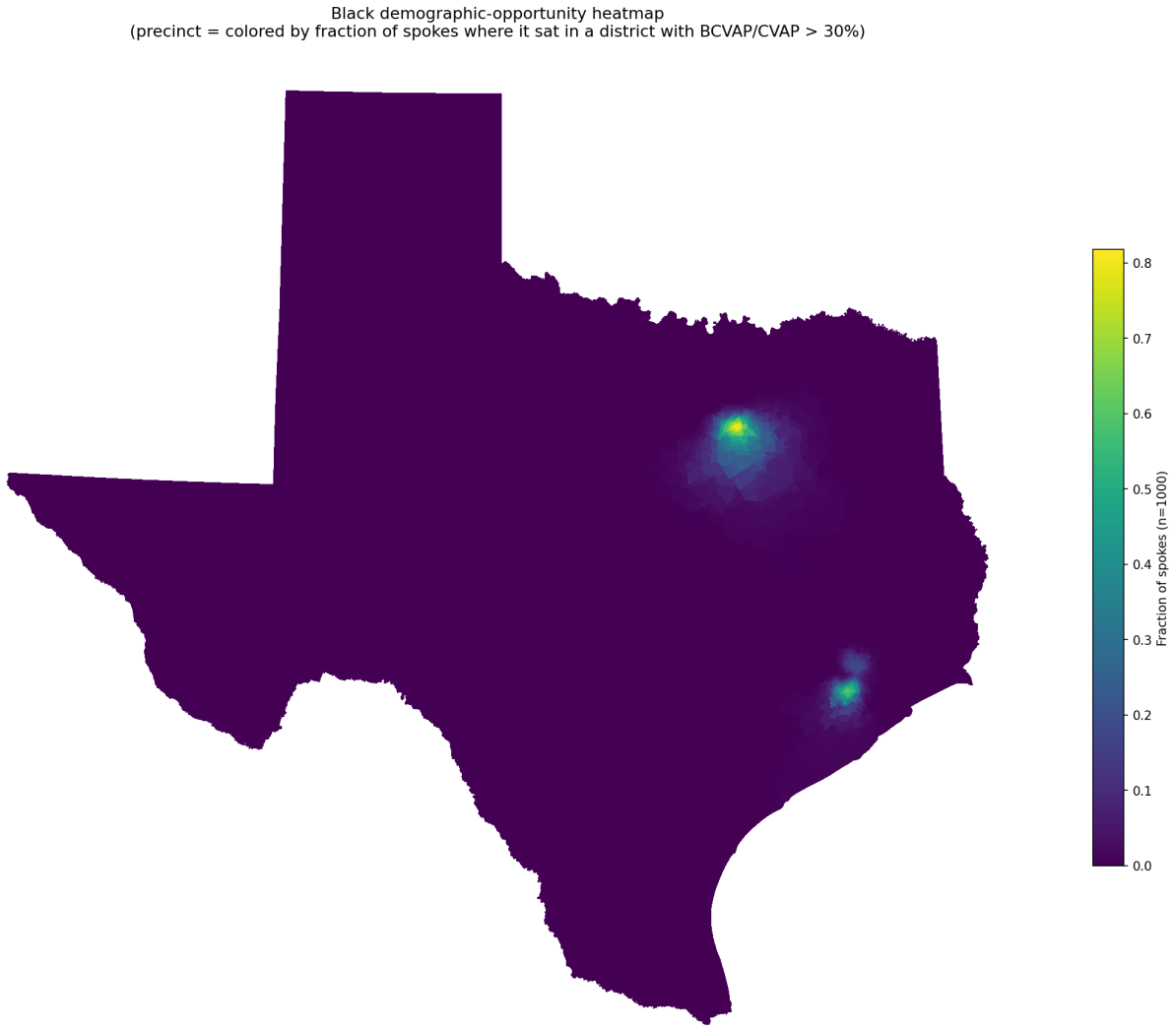
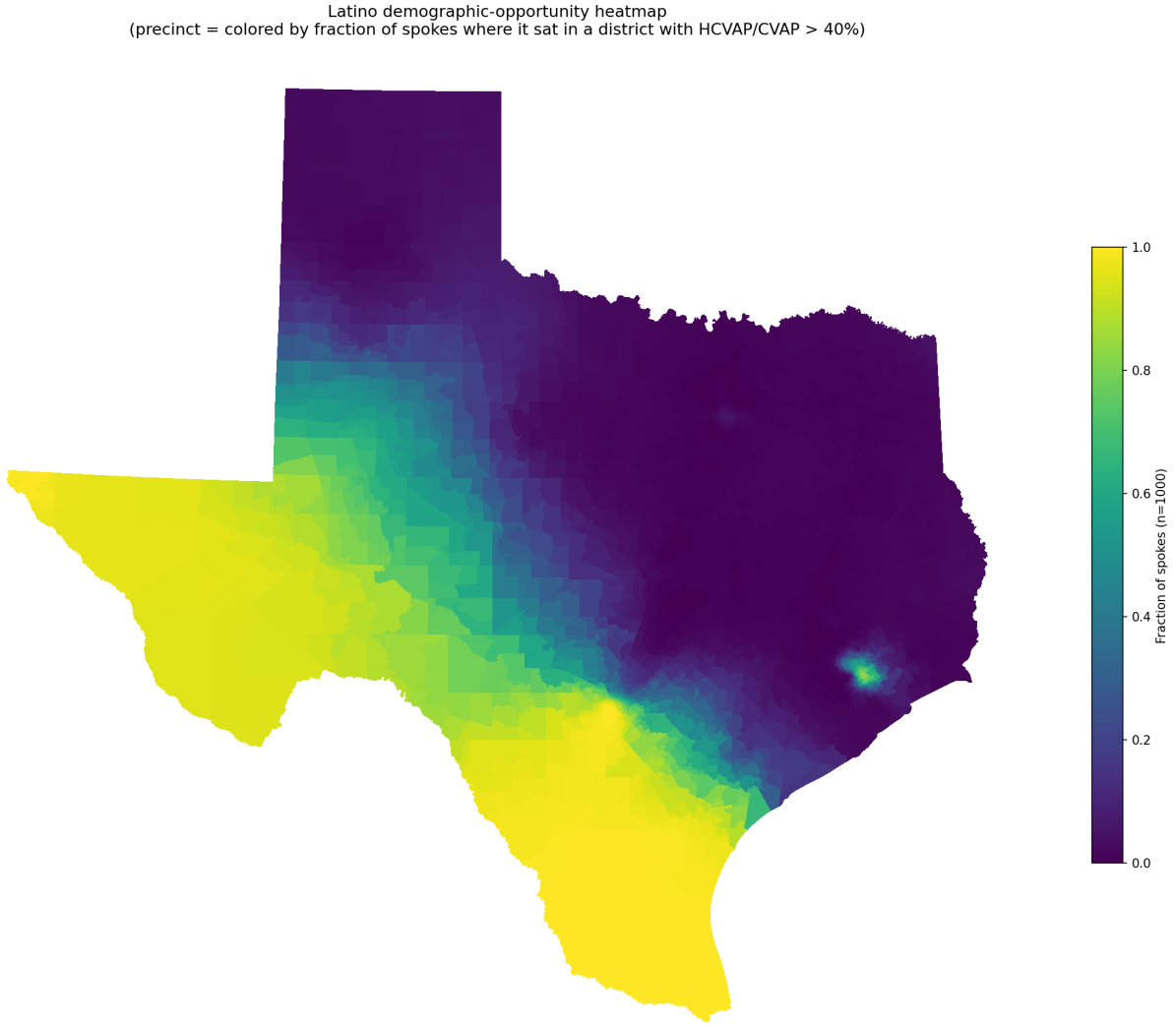
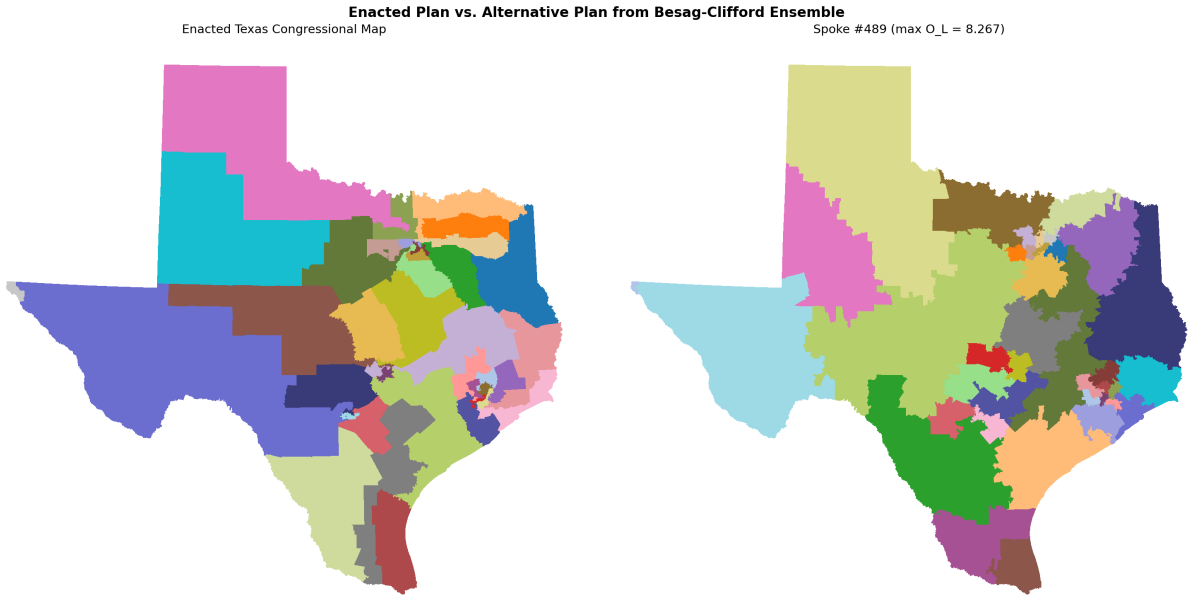


Figure 6: black opportunity heatmap


Latino demographic-opportunity heatmap
(precinct = colored by fraction of spokes where it sat in a district with HCVAP/CVAP > 40%)
1.0
0.8
0.6
0.4
0.2
0.0
Fraction of spokes (n=1000)


Enacted Plan vs. Alternative Plan from Besag-Clifford Ensemble
Enacted Texas Congressional Map
Spoke #489 (max O_L = 8.267)

Discussion

#Conclusion

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